

TiLiA – TimeLineAnnotator

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Summary

‘TiLiA’ (TimeLine Annotator) is an open-source, cross-platform network of tools and resources designed to facilitate real-time creating, displaying and interacting with timeline-style annotations of audio, video and musical scores. Developed in Python with the PySide UI framework, TiLiA is available primarily as a desktop Graphical User Interface (GUI), supported by a Command Line Interface (CLI) as well as a free and open online platform where users may upload, view and query existing analyses.

Statement of need

Visual analyses of music have proven to be an enduringly popular, effective, and versatile tool for conveying a variety of musical ideas.¹ As the famous adage goes: ‘a picture is worth a thousand words’. A comprehensive analysis of music requires being able to summarise the relationships between events that are happening concurrently and through time. Furthermore, whatever potential a *static* image has, an *interactive* one offers considerably more. Form analysis, for instance, can be hard to parse (especially in real-time) but is often easy to understand through visual summaries.

In today’s digital age, we have come to expect certain basic interactive features across visual interfaces, such as resizing content to fit the window, keyboard shortcuts, undo/redo functionality, and so on. Musical contexts extend this with additional expectations, such as real-time synchronisation (a.k.a., audio or score following).

Despite several emergent initiatives (see below), musical scholarship has not yet made the leap into this technological space with any degree of decisiveness or consistency. Even most visually minded music analyses are consigned to static images embedded in print (or in e-copies at best) and not subject to the expectations for *fair*, open-source data sharing that are now *de rigueur* in other fields.

State of the field

While one *can* appropriate score and audio editors for visual analysis (Peebles, 2013), they are limited in their annotating capabilities. Dedicated tools fall into a few groups, according to their focus. Audio-analysis tools such as Sonic Visualiser (Cannam et al., 2010) and Partiels (Guillot, 2025) centre on audio and plug-in features, and lack score integration. Score-annotation platforms such as *Edirom Online* (Röwenstrunk, 2014), *Listen Here!* (Weigl et al., 2023) and *mei-friend* (Goebel & Weigl, 2024) anchor annotations to notated elements rather than to time and lack dedicated types for analytical structures such as form, while others, such as the MPM Toolbox (Berndt, 2021), are tied to specific file formats. “Timeline-based” annotators such as

¹See Isaacson (2023) for a survey of this long history.

39 *Audio Timeliner* (Notess & Swan, 2004; Yorgason, 2018), *iAnalyse* (Couprie, 2019), *BriFormer*
40 (Jarvis, 2020) and *Dezrann* (Giraud et al., 2018) tend to focus more on providing tools for
41 rich hand-made analysis.² While each group excels at particular tasks, there remains space for
42 a solution that caters to more cross-cutting needs.

43 Platform support, for instance, is often narrow (Table 1): *Dezrann* and *BriFormer* run only
44 in the browser, where local file access is difficult and copyright media is hard to host, while
45 *iAnalyse* runs only on macOS. Few of the tools are scriptable, and most lock analyses in
46 tool-specific formats, constraining interoperability and work at scale.

47 Most software focuses on a subset of the many forms analytical annotations can take
48 (e.g. hierarchical groupings, metre, overlapping time spans, score elements, harmony, text).
49 *Audio Timeliner* and *BriFormer* allow only nested spans, whereas *Sonic Visualiser* and *iAnalyse*
50 allow overlaps, but only on a flat surface. Datasets need all of them: *SALAMI* (Smith et al.,
51 2011) and the *Wagner Ring Dataset* (Weiß et al., 2023) encode hierarchical structure, a topic
52 of active research (McFee & Ellis, 2014); the *ABC* (Neuwirth et al., 2018) and the *Annotated*
53 *Mozart Sonatas* (Hentschel et al., 2021) layer harmony, phrase and cadence labels; *BPSD*
54 (Zeitler et al., 2024) and the *Harmonix Set* (Nieto et al., 2019) align formal structure with
55 beats and measures; and the *Schubert Winterreise Dataset* (Weiß et al., 2021) links harmony
56 annotations to scores and several recordings.

57 Input formats and multimodality have varied support. Symbolic-based platforms, for instance,
58 rarely support video. The *Walküre* (Balke et al., 2017), *Freischütz Digital* (Müller et al., 2013),
59 *Erkomaishvili* (Rosenzweig et al., 2020) and *Lohengrin TimeMachine* (Lewis et al., 2021)
60 interfaces show what multimodal, synchronised and richly interactive access can offer, but are
61 tied to specific corpora.

Table 1: Comparison of music annotation tools.

Tool	Annotation type						Media					Licence	Script/Platform				
	B	C	E	H	I	O	T	A	P	S	V	Y	IO	L	M	W	w
TiLiA	●	●	○	●	●	●		●	●	●	●	●	GPL-3.0	CLI; CSV, JSON	●	●	○
Audio Timeliner				●	●		●	●					—	CLI	●	●	
BriFormer				●	●	○		●				●	—				●
Dezrann	●	●	○	●	●			●	●	●		●	GPL-3.0	JSON			●
iAnalyse			●		●	●	●	●	●	●	●		—		●		
mei-friend		●	●		●		●		●	●			AGPL-3.0	MEI			●
Sonic Visualiser	●	●			●	●	●	●					GPL-2.0	CSV, CLI	●	●	●

62 *Annotations:* **B** beat · **C** harmony · **E** score elements · **H** hierarchical · **I** instant · **O**
63 overlapping · **T** text
64 *Media:* **A** audio · **P** PDF · **S** score · **V** video · **Y** YouTube
65 *Platforms:* **L** Linux · **M** macOS · **W** Windows · **w** web
66 Hollow circles mark partial support.

67 In response to this challenge, we present ‘TiLiA’: a timeline annotator for all. It provides an
68 open-source, cross-platform desktop tool that aligns annotations with audio, video, YouTube,

²Commercial alternatives include: *Hookpad* and *Soundslice* (closed-source and subscription-based). *SyncPlayer* (Fremeroy et al., 2007; Kurth et al., 2005) was another timeline-based annotator that is no longer available.

PDFs and scores on shared timelines, offers dedicated types for musical structure, harmony
and metre, and can be scripted and exchanged through CSV and JSON.

71 Specifications

72 TiLiA consists of an organisation of digital tools, primarily of its [open-source desktop application](#)
73 [and command-line interface](#), but also of a supporting online platform. The desktop application
74 is developed in Python using the PySide binding for the Qt UI framework. Automated testing
75 is done with pytest, and fully-automated deployment is available via GitHub Actions. The
76 code base is loosely organised around a Model-View-Controller (MVC) pattern, which allows
77 both the CLI and the GUI to benefit from the same backend logic. We provide object-oriented
78 base classes such as Timeline, TimelineUI and TimelineComponent as a means to promote
79 extensibility and support an evergrowing range of annotation needs.

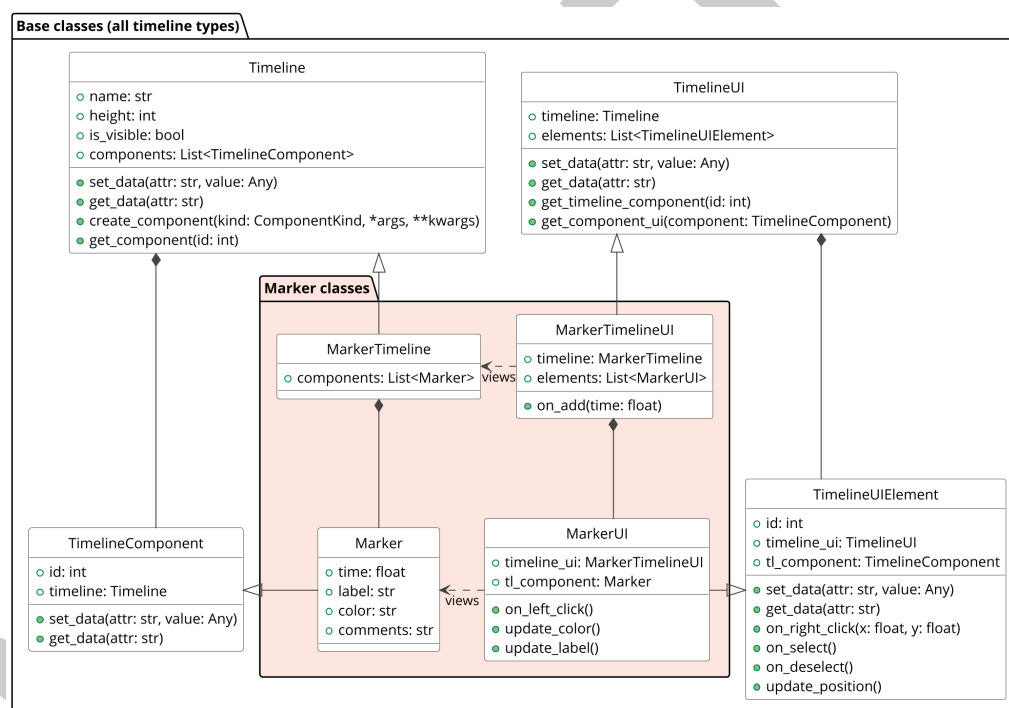


Figure 1: A *simplified* diagram for the main timeline base classes.

Figure 1 is a *simplified* diagram for the main timeline base classes. The outer box shows the base classes shared by all timeline types; the inner box shows the concrete subclasses for the marker timeline type, one example among the various timeline types. Timeline acts as container for TimelineComponents. Both act as models of TimelineUI and TimelineUIElement, which are views; in the marker example, MarkerTimelineUI and MarkerUI view MarkerTimeline and Marker respectively.

TiLiA is designed both for the manual creation of analyses — particularly where this requires complex alignment with audio/video sources — and for the large-scale conversion and import of external analyses. While the `.tla` format may still change as functionality grows, documentation is available for [importing](#) from CSV and [exporting](#) to JSON: both steps can be fully automated through the CLI. The [tilia-dcml](#) repository demonstrates this by integrating an external corpus of musical analyses.

92 **The desktop application**

93 The main part of TiLiA is a cross-platform *What-You-See-Is-What-You-Get* GUI (Figure 2),
94 designed to feel intuitive to anyone who has worked with audiovisual editing software. A
95 TiLiA analysis consists of timelines, each containing annotations that can be aligned to an
96 audio, video, YouTube stream or musical score. Each *timeline type* is designed to support
97 conventional formal annotations, with the affordance to work with a variety of other analytical
98 styles. These timelines each contain one or more *component types* (e.g., markers for marker
99 timelines), each with different *properties* to convey information including comments, colour and
100 and labels.

101 We connect timelines and their components by using the loaded media as a “base layer” to
102 which all timelines refer. Timeline components have either time or start_time and end_time
103 timestamps which position them in relation to the underlying media. If a different media is
104 loaded, the user has the option to either uniformly scale timelines to the new media or to keep
105 them at their original position (deleting out-of-bounds components if necessary). If a beat
106 timeline is present, the software can map timestamps to metric positions, also providing a
107 metric-based interface for importing, exporting and interacting with timelines.

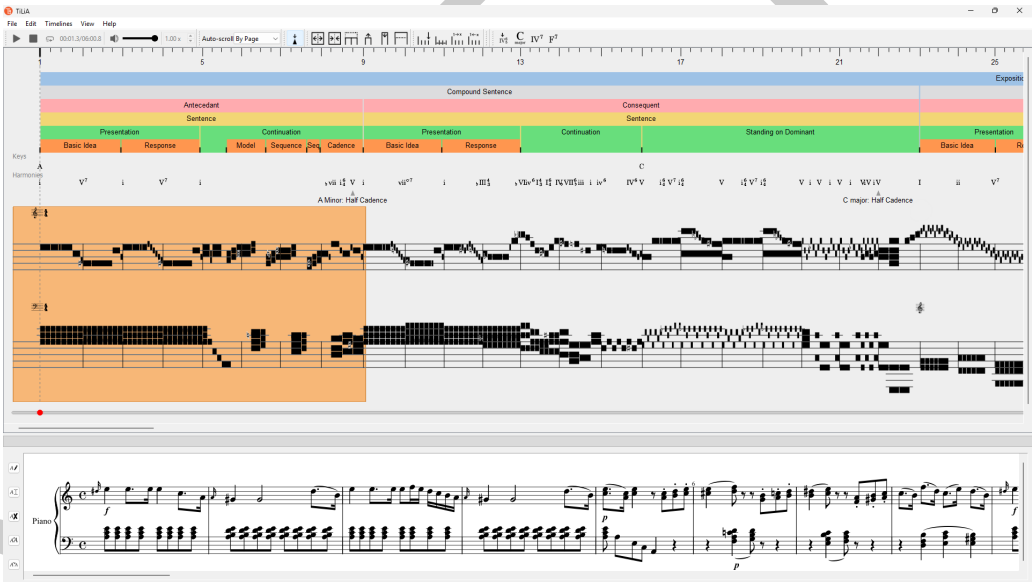


Figure 2: Excerpt of a TiLiA analysis on the desktop application.

108 Currently, there are eight types of timelines:

Timeline type	Purpose
Audiowave	Displaying amplitude graphs
Beat	Marking beats/measures, synchronising metrical and absolute time
Harmony	Chord symbols and roman numeral analysis
Hierarchy	Representing hierarchical relations and sequences
Marker	Annotating discrete timepoints
PDF	Synchronising external PDF documents with playback
Range	Representing non-hierarchical relations and sequences
Score	Synchronising digitised scores with playback

Command-line interface

In addition to the desktop GUI and the web app, a **command line interface** allows the automation of file and timeline creation and edition, media loading and external data importing. Sequences of commands can be loaded by the CLI from scripts written in plain text files, enabling programmatic conversion from various data formats to TiLiA files.

```
metadata set-media-length 120
timelines add hierarchy --name "Form"
timelines add marker --name "Formal closures"
timelines add hierarchy --name "Harmonic segments"
timelines add beat --name "Measures" --beat-pattern 1
timelines add marker --name "Harmony"
timelines add marker --name "Harm. functions"
timelines add marker --name "Keys"

load-media <path to audio> --scale-timelines yes
metadata set title "Title"
save <path to TiLiA file> --overwrite
export <path to JSON file> --overwrite
```

Online platform

The TiLiA desktop application is supported by an [online platform](#) that lets users store, share, visualise and even query existing analyses. Registration is free and is only required for uploading content. This reflects a goal of TiLiA to support not only the modernised creation and storage of analyses, but also to support the still-rare practice of analytical *collaboration* and structured *searches* on the corpora; both which benefit from network effects facilitated by the lowered entry barrier.

Outlook

The TiLiA desktop application aims to provide a user-friendly annotation tool both for MIR researchers and musicologists. It can be used to create large annotation datasets or detailed, interactive analyses of individual pieces. It can also be used in the classroom to better connect theoretical concepts with audiovisual material or as a common interface for analytical assignments.

In the future we hope to convert existing open-source datasets and integrate them into the web platform, providing a centralised hub where diversified audiovisual data can be queried and interacted with in a uniform manner. We also plan to improve audio processing and score displaying capabilities and to add more timeline types (e.g. a 2-dimensional graph timeline) to offer more diversified visualisations.

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